

Homework 8, due Oct 29 before class

1. Charged particle (charge e mass m) motion in circularly polarized EM wave propagating along magnetic field.

A circularly wave of amplitude $E_w = B_w$ propagates along the guiding magnetic field B_0 . Find velocity of the particle v (non-relativistic). Keep in mind that there are two possible circular polarizations, two possible signs of charge and two possible directions of wave propagating (along and against the guide field).

Hint.

Let the guide field be along z . Write electromagnetic field of the wave (check that Maxwell's equations are satisfied). Assume that at each moment $v_z = 0$. Let a particle be located at $z = 0$.

Assume that at each moment velocity is aligned or counter-aligned with the waves' magnetic field, and hence perpendicular to the electric field (energy and value of velocity are conserved).